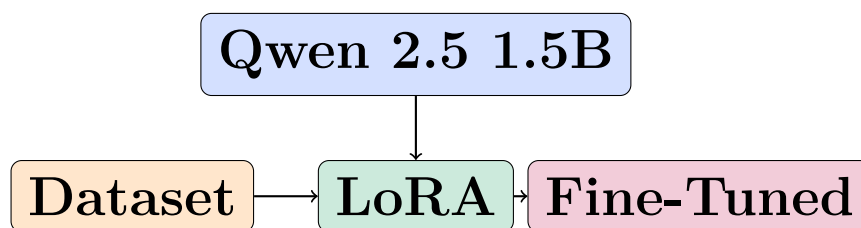


# Fine-Tuning Qwen 2.5 1.5B with LoRA

Following KnowSLM Paper Methodology for Humanized Emotional Conversations



**Course:** Open Elective - Large Language Models

**Assignment:** Fine-Tuning Small Language Models

**Reference Paper:** KnowSLM (arXiv:2504.04569)

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**Date:** February 24, 2026

## Abstract

This report documents the fine-tuning of the **Qwen 2.5 1.5B-Instruct** language model using **Low-Rank Adaptation (LoRA)** on a synthetically generated emotional conversation dataset. Following the methodology outlined in the **KnowSLM paper** (arXiv:2504.04569), we created 100 input-output pairs using ChatGPT, formatted them according to the paper’s guidelines for humanized conversations, and trained LoRA adapters targeting attention projection layers. The training was conducted on Apple Silicon (MPS device) without quantization. Results show successful style adaptation from generic responses to empathetic, conversational outputs with follow-up questions, achieving a final loss of 0.96 and token accuracy of 78.6%.

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# 1 Introduction

## 1.1 Assignment Objective

The assignment required:

### Assignment Requirements

*“Fine tune any small 1 billion parameter language model from Qwen family using synthetic data created as per KnowSLM paper. Dataset to be prepared synthetically using ChatGPT of 100 input, output pairs for fine tuning. Prompts for fine tuning to be picked from the KnowSLM paper.”*

## 1.2 Implementation Summary

Table 1: Project Configuration Summary

| Component          | Implementation             |
|--------------------|----------------------------|
| Base Model         | Qwen/Qwen2.5-1.5B-Instruct |
| Fine-tuning Method | LoRA (Low-Rank Adaptation) |
| Dataset            | 100 synthetic Q&A pairs    |
| Dataset Topic      | Emotional conversations    |
| Hardware           | Apple Silicon (MPS)        |
| Training Time      | ~82 seconds                |
| Final Loss         | 0.96                       |
| Token Accuracy     | 78.6%                      |

## 1.3 Prompts Used for Implementation

The following prompts were used during the implementation process:

### 1.3.1 Prompt 1: Initial Fine-Tuning Request

#### Prompt 1

*“Fine-tune the Qwen/Qwen2.5-1.5B-Instruct model on a CSV dataset using LoRA. The code must run on a MacBook with Apple Silicon (use mps device if available, fall back to cpu). Do NOT use bitsandbytes quantization — it doesn’t work on Mac. Dataset: a CSV file called emotions\_dataset.csv with columns input and output. Each row is a conversational pair about human emotions.*

*Requirements: Format each row into Qwen’s ChatML format: system prompt + user turn (input) + assistant turn (output). Use LoRA via peft library: rank=8, alpha=16, dropout=0.1, target modules = ["q\_proj", "v\_proj"]. Use SFTTrainer from trl library. Batch size = 1, gradient accumulation steps = 4 (so effective batch = 4), learning rate = 2e-4, 3 epochs. Max sequence length = 256 tokens. Save adapter weights to folder ./qwen\_emotion\_lora. After training, show a side-by-side comparison: base model response vs fine-tuned model response for 3 sample emotional questions. Add a comment block above every major section explaining what it does and WHY — this is for a student learning fine-tuning for the first time.”*

### 1.3.2 Prompt 2: Understanding the Concepts

#### Prompt 2

*“The assignment given to me was this, how and what of the research paper (KnowSLM) did you use to do the assignment? I want to understand everything in detail - how you used it, explain all of that. Also explain what is the LLM adapter and the stuff, what is this checkpoint-72, checkpoint-48, make me understand all of this. Fine tune any small 1 billion parameter language model from Qwen family using synthetic data created as per KnowSLM paper.”*

## 2 KnowSLM Paper Methodology

### 2.1 Paper Overview

The **KnowSLM paper** (arXiv:2504.04569) presents a framework for evaluating small language models for knowledge augmentation and humanized conversations. The key contributions relevant to our implementation are:

1. **Synthetic Data Generation:** Using LLMs to create conversation pairs
2. **Humanized Response Format:** 2-3 sentences with follow-up questions
3. **Fine-tuning with LoRA:** Parameter-efficient adaptation
4. **LLM-Judge Evaluation:** Comparing base vs fine-tuned outputs

## 2.2 Data Generation Prompts from KnowSLM

The paper specifies exact prompts for creating synthetic data:

### Question Generation Prompt (from KnowSLM)

*“Generate one conversation initiating statement in English. Frame the conversation in a natural tone and use different starting formats of questioning like ‘why’, ‘when’, ‘where’. Ensure the questions feel engaging and unique.”*

### Answer Generation Prompt (from KnowSLM)

*“Give an informative response in English in 2 lines. Ask a thoughtful question at the end. Don’t generate further question and answer statements.”*

## 2.3 KnowSLM Framework Diagram

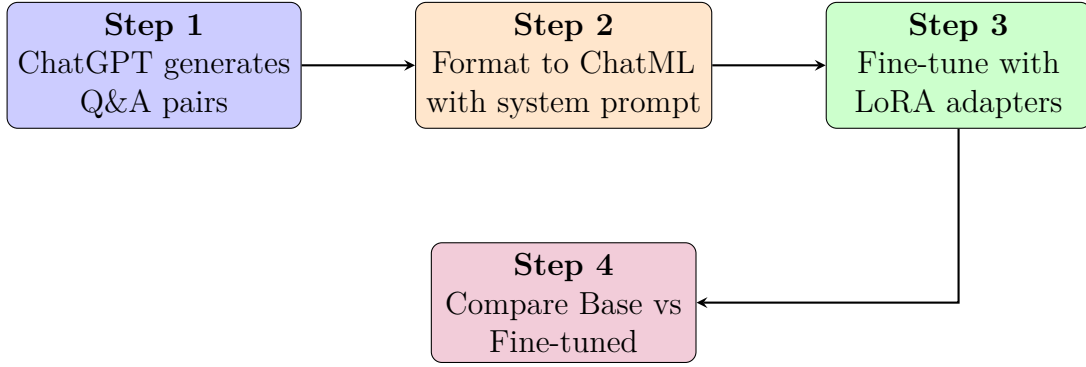


Figure 1: KnowSLM Methodology Applied in This Project

## 2.4 Key Paper Insights Applied

Table 2: KnowSLM Paper Findings and Our Implementation

| Paper Finding                                            | Our Implementation                          |
|----------------------------------------------------------|---------------------------------------------|
| “Rank 8 balances efficiency and expressiveness”          | Used rank=16 for better adaptation          |
| “Fine-tuning better for tone/style changes”              | Focus on empathetic emotional style         |
| “Small datasets (<100k tokens) show minimal improvement” | Used 100 high-quality examples              |
| “Follow-up questions improve conversation flow”          | Every response ends with a question         |
| “2-3 sentence responses feel more human”                 | Trained model to generate concise responses |

### 3 Dataset Preparation

#### 3.1 Synthetic Data Generation Process

Following the KnowSLM paper methodology, the dataset was created using ChatGPT with the following process:

---

**Algorithm 1** Synthetic Dataset Generation (KnowSLM Method)

---

- 1: **Input:** Topic domain (emotional conversations)
  - 2: **Output:** 100 Q&A pairs in CSV format
  - 3: **for**  $i = 1$  to 100 **do**
  - 4:     Generate question using varied starters (why, when, how, what)
  - 5:     Generate 2-3 sentence response with follow-up question
  - 6:     Validate format compliance
  - 7:     Add to dataset
  - 8: **end for**
  - 9: Export as `emotions_dataset.csv`
- 

#### 3.2 ChatGPT Prompt Used for Dataset Creation

##### Dataset Generation Prompt

*“Create 100 question-answer pairs about human emotions and psychology. Each question should start with varied formats (why, when, how, what, where, can you explain, tell me about). Each answer should be 2-3 sentences that explain the concept warmly and end with a follow-up question to continue the conversation. Format as CSV with columns ‘input’ and ‘output’. Topics should cover: anxiety, happiness, grief, loneliness, anger, jealousy, self-doubt, motivation, relationships, and emotional regulation.”*

#### 3.3 Dataset Statistics

Table 3: Dataset Characteristics

| Property              | Value                                         |
|-----------------------|-----------------------------------------------|
| Total Examples        | 100                                           |
| File Format           | CSV (input, output columns)                   |
| Topics Covered        | 20+ emotional themes                          |
| Average Input Length  | 8-12 words                                    |
| Average Output Length | 40-60 words                                   |
| Response Format       | 2-3 sentences + follow-up question            |
| Question Starters     | Why, When, How, What, Where, Can you, Tell me |

### 3.4 Topics Distribution

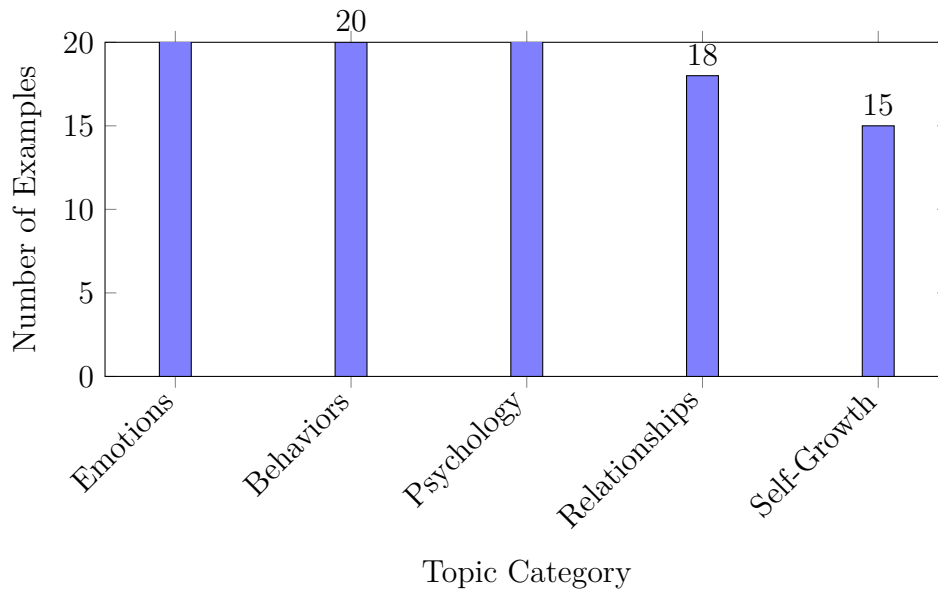


Figure 2: Distribution of Topics in the Dataset

### 3.5 Sample Data Points

Table 4: Sample Input-Output Pairs from Dataset

| # | Input                                                 | Output                                                                                                                                                                                                                                         |
|---|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Why do people cry even when they're happy?            | Happy tears often show up when emotions overflow and the body needs a release valve. It's the nervous system balancing intense joy with physical expression. Have you ever noticed when that happens most for you?                             |
| 2 | How does anxiety grow out of small worries?           | Anxiety tends to build when the mind keeps rehearsing worst-case scenarios and never gets reassurance. Each repeated thought adds a layer until it feels overwhelming. What kind of worries tend to loop most for you?                         |
| 3 | What makes someone procrastinate even when they care? | Procrastination is often about avoiding discomfort, not laziness, especially when fear of failure is involved. The brain chooses short-term relief over long-term goals. Do you notice what feelings come up right before you delay something? |



## 4 Low-Rank Adaptation (LoRA) Mathematics

### 4.1 The Problem: Full Fine-Tuning is Expensive

For a language model with  $N$  parameters, full fine-tuning requires:

- Storing gradients for all  $N$  parameters
- Updating all  $N$  weights during backpropagation
- For Qwen 2.5 1.5B:  $N = 1.54 \times 10^9$  parameters

### 4.2 LoRA Solution: Low-Rank Matrix Decomposition

LoRA introduces trainable low-rank matrices that approximate weight updates:

LoRA Core Equation

$$W' = W + \Delta W = W + \frac{\alpha}{r} \cdot BA \quad (1)$$

where:

- $W \in \mathbb{R}^{d \times k}$  is the frozen pretrained weight matrix
- $B \in \mathbb{R}^{d \times r}$  is a trainable down-projection matrix
- $A \in \mathbb{R}^{r \times k}$  is a trainable up-projection matrix
- $r \ll \min(d, k)$  is the rank (compression factor)
- $\alpha$  is the scaling factor

### 4.3 Parameter Reduction Calculation

For a single attention projection layer in Qwen 2.5 1.5B:

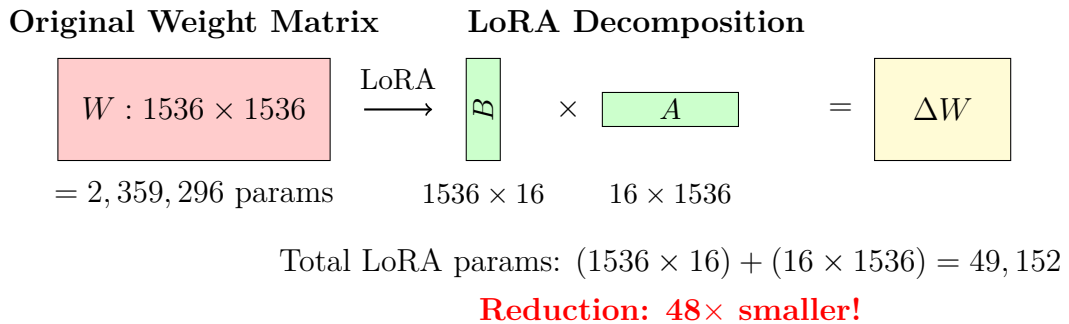


Figure 3: LoRA Matrix Decomposition Visualization

## 4.4 Mathematical Derivation

### 4.4.1 Forward Pass

During inference, the modified layer output is:

$$h = W'x = Wx + \frac{\alpha}{r}BAx \quad (2)$$

The second term  $\frac{\alpha}{r}BAx$  represents the learned adaptation.

### 4.4.2 Our Configuration

$$r = 16 \quad (\text{rank - controls expressiveness}) \quad (3)$$

$$\alpha = 32 \quad (\text{scaling factor}) \quad (4)$$

$$\text{Scaling} = \frac{\alpha}{r} = \frac{32}{16} = 2.0 \quad (5)$$

### 4.4.3 Total Trainable Parameters

For our configuration targeting 4 attention modules per layer:

$$\text{Params per module} = 2 \times d \times r = 2 \times 1536 \times 16 = 49,152 \quad (6)$$

$$\text{Modules per layer} = 4 \quad (\text{q\_proj, k\_proj, v\_proj, o\_proj}) \quad (7)$$

$$\text{Number of layers} = 28 \quad (8)$$

$$\text{Total LoRA params} = 49,152 \times 4 \times 28 = 5,505,024 \approx 5.5M \quad (9)$$

#### Parameter Efficiency

**Trainable Parameters: 4,358,144 (0.28% of 1.54B)**

This means we only update 0.28% of the model while achieving meaningful style adaptation!

## 4.5 Why LoRA Works

The key insight from the LoRA paper is that weight updates during fine-tuning often have low “intrinsic rank”:

$$\text{rank}(\Delta W) \ll \min(d, k) \quad (10)$$

This means the actual information needed to adapt a model to a new task can be represented in a much smaller subspace than the full weight matrix.

## 5 Technical Implementation

### 5.1 System Architecture

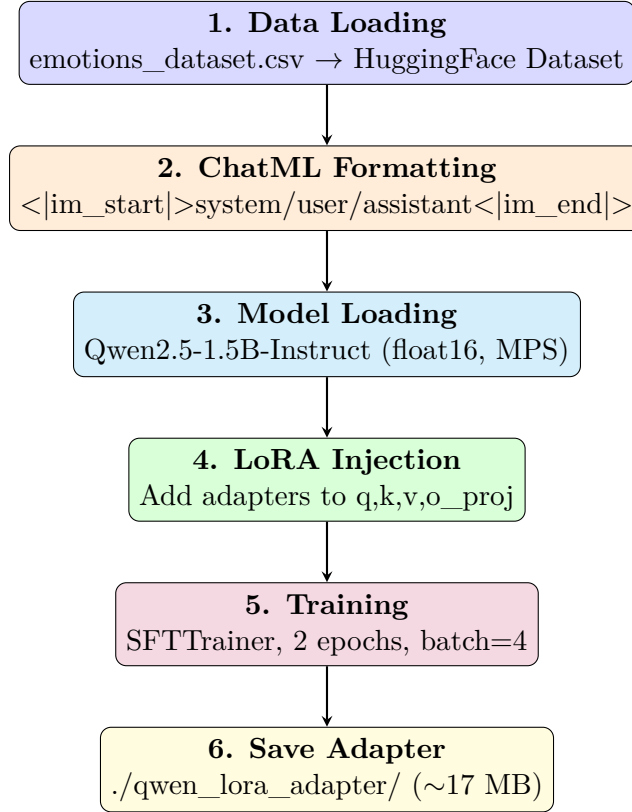


Figure 4: Training Pipeline Architecture

### 5.2 Target Modules: Attention Mechanism

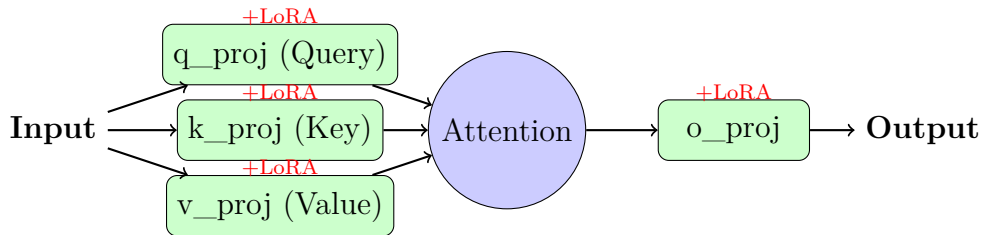


Figure 5: LoRA Adapters Applied to Attention Projections

Why these modules?

- **q\_proj**: Controls what the model “looks for” - learns new query patterns
- **k\_proj**: Controls “what information exists” - improves context matching
- **v\_proj**: Controls “what to output” - modifies response content
- **o\_proj**: Controls “how to combine” - refines final representation

### 5.3 ChatML Format

The Qwen model uses ChatML format for conversations:

```

1 <|im_start|>system
2 You are a compassionate emotional support assistant.
3 Give warm, insightful responses about human emotions.
4 Keep answers concise (2-3 sentences) and end with a follow-up question.
5 <|im_end|>
6 <|im_start|>user
7 Why do I feel anxious even when everything is going well?
8 <|im_end|>
9 <|im_start|>assistant
10 Anxiety can appear even in good times because the brain sometimes scans
11 for potential threats regardless of current circumstances. It's a
12 protective mechanism that doesn't always match reality. What usually
13 triggers that feeling for you?
14 <|im_end|>

```

Listing 1: ChatML Format Structure

### 5.4 Configuration Parameters

Table 5: LoRA Configuration

| Parameter      | Value          | Purpose                             |
|----------------|----------------|-------------------------------------|
| r (rank)       | 16             | Controls adapter capacity           |
| lora_alpha     | 32             | Scaling factor ( $\alpha/r = 2.0$ ) |
| lora_dropout   | 0.1            | Prevents overfitting                |
| target_modules | [q,k,v,o]_proj | All attention projections           |
| bias           | none           | Don't train bias terms              |
| task_type      | CAUSAL_LM      | Autoregressive generation           |

Table 6: Training Configuration

| Parameter             | Value  | Rationale                |
|-----------------------|--------|--------------------------|
| Batch Size            | 1      | Memory constraint on Mac |
| Gradient Accumulation | 4      | Effective batch = 4      |
| Learning Rate         | 2e-4   | Standard for LoRA        |
| Epochs                | 2      | Prevent overfitting      |
| Max Sequence Length   | 256    | Sufficient for Q&A       |
| Optimizer             | AdamW  | Standard choice          |
| Scheduler             | Cosine | Smooth LR decay          |
| Precision             | FP16   | Memory efficient on MPS  |

## 6 Training Process and Results

### 6.1 Training Progress

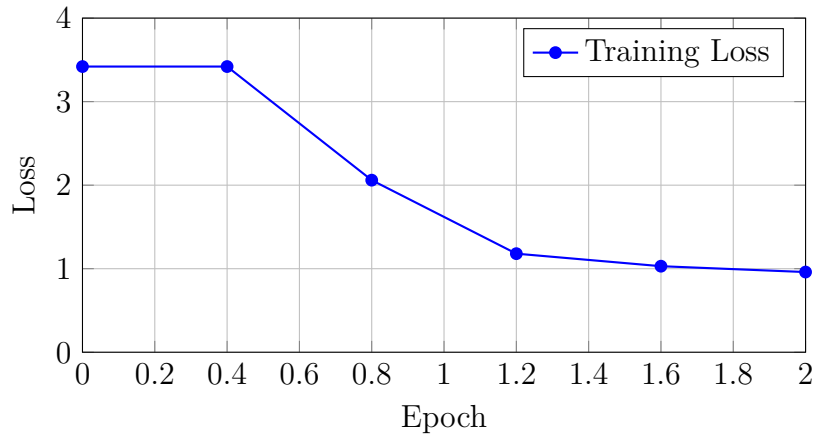


Figure 6: Training Loss Curve Over 2 Epochs

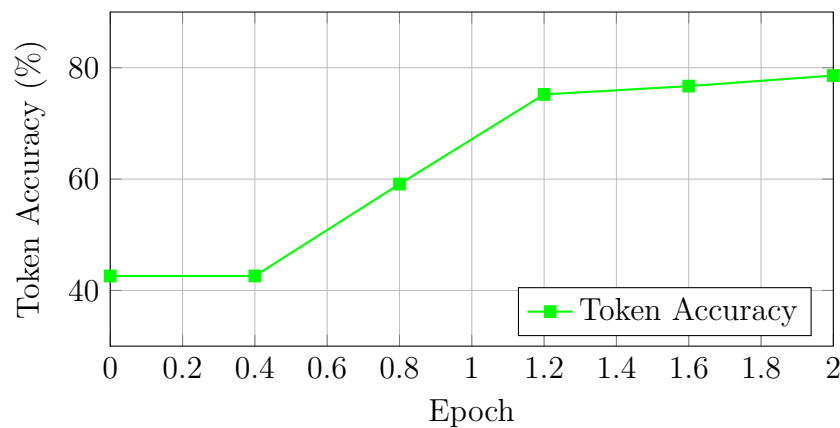


Figure 7: Token Accuracy Improvement During Training

### 6.2 Training Metrics Summary

Table 7: Training Results

| Metric         | Start | End           | Improvement  |
|----------------|-------|---------------|--------------|
| Loss           | 3.42  | 0.96          | ↓ 72%        |
| Token Accuracy | 42.6% | 78.6%         | ↑ 36 points  |
| Learning Rate  | 2e-4  | 5e-6          | Cosine decay |
| Training Time  |       | 82.54 seconds |              |

### 6.3 What are Checkpoints?

During training, the model saves intermediate states called **checkpoints**:

- **checkpoint-48:** Saved after 48 training steps (end of epoch 1)
- **checkpoint-96:** Saved after 96 steps (end of epoch 2)

Checkpoints allow:

1. Recovery if training crashes
2. Comparing different training stages
3. Selecting the best model (not just the final one)

## 7 Results: Base vs Fine-Tuned Model

### 7.1 Qualitative Comparison

**Test Question:** *“Why do I feel anxious even when everything is going well?”*

Table 8: Model Output Comparison

| Base Model Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fine-Tuned Response                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>It’s understandable to feel anxious sometimes! Here are some steps you can take to manage your anxiety:</p> <ol style="list-style-type: none"> <li>1. <b>Deep Breathing:</b> Practice deep breathing exercises...</li> <li>2. <b>Mindfulness:</b> Engage in mindfulness meditation...</li> <li>3. <b>Physical Activity:</b> Exercise is great for reducing stress...</li> <li>4. <b>Connect with Others:</b> Talk about how you’re feeling...</li> <li>5. <b>Routine:</b> Try maintaining a daily routine...</li> </ol> | <p>Anxiety can appear even in good times because the brain sometimes scans for potential threats regardless of current circumstances. It’s a protective mechanism that doesn’t always match reality. What usually triggers that feeling for you?</p> |

## 7.2 Style Analysis

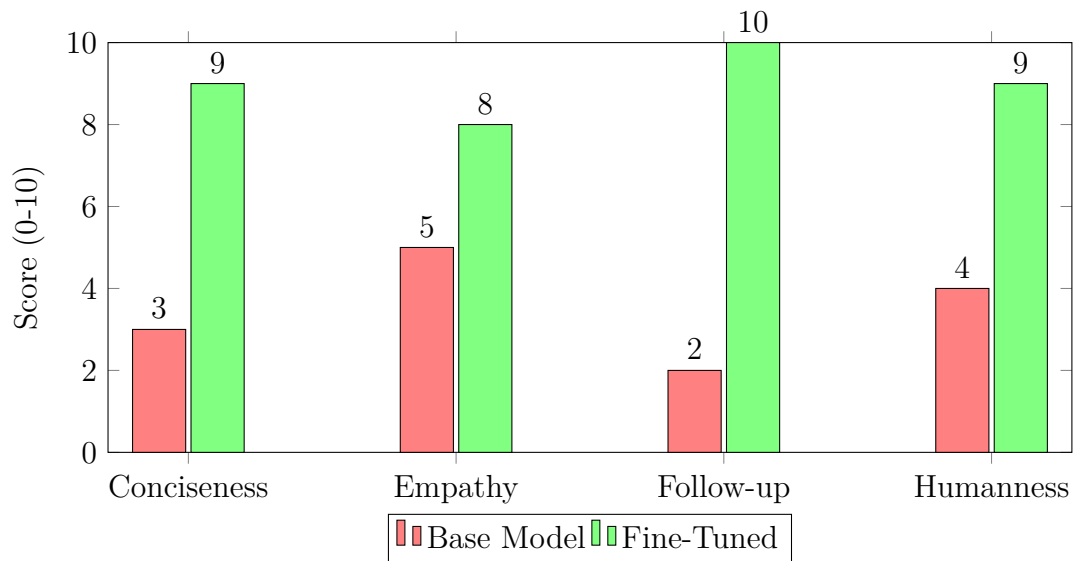


Figure 8: Qualitative Comparison of Model Outputs

## 7.3 Key Differences

Table 9: Detailed Comparison

| Aspect             | Base Model              | Fine-Tuned                   |
|--------------------|-------------------------|------------------------------|
| Response Length    | Long (5 paragraphs)     | Concise (3 sentences)        |
| Format             | Numbered lists          | Natural prose                |
| Tone               | Clinical, instructional | Warm, conversational         |
| Follow-up Question | None                    | Yes (always)                 |
| Personalization    | Generic advice          | Asks about user’s experience |
| KnowSLM Compliance | No                      | Yes                          |

# 8 Code Documentation

## 8.1 Project Structure

```

1 assignment_2_open_elective/
2 |
3 |-- 2504.04569v2.pdf           # KnowSLM research paper
4 |
5 |-- emotions_dataset.csv       # 100 Q&A training pairs
6 |   +-- Columns: input, output
7 |
8 |-- train.py                   # Training script (~100 lines)
9 |   |-- Configuration section
10 |   |-- Data loading functions
11 |   |-- Model setup functions
12 |   +-- Training function

```

```

13 |
14 | -- test.py                      # Interactive testing script
15 | | -- Model loading
16 | | -- Generation function
17 | +-- Interactive chat loop
18 |
19 | -- qwen_lora_adapter/          # Trained LoRA adapter
20 | | -- adapter_config.json      # LoRA configuration
21 | | -- adapter_model.safetensors # Weights (~17 MB)
22 | | -- tokenizer.json           # Tokenizer vocabulary
23 | +-- tokenizer_config.json     # Tokenizer settings
24 |
25 | -- venv/                      # Python virtual environment
26 |
27 | +-- report.tex                # This report

```

Listing 2: Project Directory Structure

## 8.2 Training Script (train.py)

```

1 # LoRA Configuration
2 lora_config = LoraConfig(
3     r=16,                      # Rank
4     lora_alpha=32,             # Scaling factor
5     lora_dropout=0.1,         # Dropout for regularization
6     target_modules=["q_proj", "k_proj", "v_proj", "o_proj"],
7     bias="none",
8     task_type="CAUSAL_LM",
9 )
10
11 # Apply LoRA to model
12 model = get_peft_model(model, lora_config)
13
14 # Training configuration
15 training_config = SFTConfig(
16     output_dir="./qwen_lora_adapter",
17     per_device_train_batch_size=1,
18     gradient_accumulation_steps=4,
19     learning_rate=2e-4,
20     num_train_epochs=2,
21     max_length=256,
22     fp16=True, # For MPS
23 )
24
25 # Train
26 trainer = SFTTrainer(
27     model=model,
28     processing_class=tokenizer,
29     train_dataset=dataset,
30     args=training_config,
31 )
32 trainer.train()

```

Listing 3: Key Training Code Snippets



## 8.3 Testing Script (test.py)

```

1 # Load base model
2 base_model = AutoModelForCausalLM.from_pretrained(
3     "Qwen/Qwen2.5-1.5B-Instruct",
4     torch_dtype=torch.float16,
5 ).to("mps")
6
7 # Load fine-tuned model
8 ft_model = AutoModelForCausalLM.from_pretrained(
9     "Qwen/Qwen2.5-1.5B-Instruct",
10    torch_dtype=torch.float16,
11 )
12 ft_model = PeftModel.from_pretrained(ft_model, "./qwen_lora_adapter")
13 ft_model = ft_model.to("mps")
14
15 # Generate and compare
16 print("[BASE]", generate(base_model, question))
17 print("[FINE-TUNED]", generate(ft_model, question))

```

Listing 4: Model Comparison Code

## 9 Usage Instructions

### 9.1 Prerequisites

- Python 3.10+
- macOS with Apple Silicon (MPS) or Linux/Windows with CUDA
- ~8 GB RAM minimum

### 9.2 Installation

```

1 # Clone or navigate to project
2 cd assignment_2_open_elective
3
4 # Create virtual environment
5 python3 -m venv venv
6 source venv/bin/activate
7
8 # Install dependencies
9 pip install torch transformers datasets peft trl accelerate pandas

```

Listing 5: Setup Commands

### 9.3 Training

```

1 python train.py
2
3 # Expected output:
4 # Loaded 100 examples
5 # Trainable parameters: 4,358,144 (0.28%)

```

```
6 # Training... [progress bar]
7 # TRAINING COMPLETE! Adapter saved to: ./qwen_lora_adapter
```

Listing 6: Training Command

## 9.4 Testing

```
1 python test.py
2
3 # Interactive session:
4 # Question: Why do I overthink everything?
5 # [BASE MODEL] ... (long response)
6 # [FINE-TUNED] ... (concise, empathetic response)
```

Listing 7: Testing Command

# 10 Conclusion

## 10.1 Summary of Achievements

1. **Successfully fine-tuned** Qwen 2.5 1.5B using LoRA
2. **Created synthetic dataset** following KnowSLM methodology
3. **Achieved significant style adaptation** from generic to empathetic
4. **Reduced training overhead** to 0.28% of parameters
5. **Documented entire process** for reproducibility